

SEG3125 – Portfolio Website Report

Assignment 1

SEG3125 - Analysis & Design of User Interface

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School of Engineering and Computer Science

University of Ottawa

Course Coordinator: Caroline Barrière

Ajay Suthaharan - 300351570

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1. Introduction

This project involved designing and developing a personal portfolio website utilizing frontend technologies including React, HTML, CSS, etc. The goal of the website was to present personal information, technical experience, featured projects, and UI/UX coursework in a clean, professional, and visually structured format. The portfolio was designed to function both as a course deliverable and as a professional portfolio that can continue to be used for internships, co-op applications, and software development opportunities.

The final website includes sections for personal background, technical skills, professional experience, featured software projects, course design work, and contact information. It was deployed publicly using Netlify and designed with responsiveness, usability, and visual consistency in mind.

2. Previous Experience with JavaScript & React

Before this assignment, I had prior experience working with JavaScript through both academic and personal software projects. I used JavaScript in full-stack and frontend development environments, including building interactive user interfaces, working with APIs, and implementing application logic.

During a software development internship, I contributed to frontend interface development and software workflows involving JavaScript-based technologies. In addition, I gained experience through personal projects and hackathons, where I worked with React and modern frontend tooling to create interactive applications and dashboards. For example, I developed projects using React and Vite that involved dynamic user interfaces, analytics visualizations, routing, and real-time interactions.

Although I had prior exposure to JavaScript and React, this assignment provided an opportunity to further improve my understanding of component-based design, reusable UI structures, responsive layouts, state organization, and frontend styling using Tailwind CSS. It also helped reinforce best practices for structuring a professional and maintainable React application.

3. Tutorials, Resources, and Readings Used

Although I had prior experience with JavaScript and React development, several resources were used throughout the assignment to reinforce implementation details, improve UI quality, and troubleshoot specific issues.

Official documentation was frequently referenced, particularly React documentation for component structure, reusable UI organization, routing, and rendering concepts. Tailwind CSS documentation was also used extensively to understand utility classes, responsive layouts, spacing systems, hover effects, and styling consistency.

In addition, online technical references and developer resources were consulted for React Router integration, Netlify deployment, smooth scrolling behavior, navigation handling, and responsive UI patterns. General frontend development resources and examples were also used to refine visual hierarchy, spacing, and card-based layout design.

These resources were primarily used to supplement existing experience and support implementation decisions rather than to learn JavaScript or React from the beginning.

4. Portfolio Sites that Inspired the Design

Several portfolio websites influenced the visual direction, layout, and interaction choices of this project.

The portfolio of **Brittany Chiang** (<https://brittanychiang.com/>) inspired the clean and professional presentation of technical information. The website uses strong visual hierarchy, clear section organization, and effective spacing to prioritize readability and make projects and experience easy to scan. This influenced my decision to structure information into clearly separated sections while maintaining a clean and professional visual style.

The portfolio of **Ryan Mulligan** (<https://ryanmulligan.dev/>) influenced the presentation of projects and the use of modern visual styling. The site demonstrates consistent typography, balanced spacing, and a polished card-based structure that presents technical content in an organized way. This inspired aspects of my portfolio layout, including structured project cards, hover interactions, and consistent visual grouping.

The portfolio of **Bruno Simon** (<https://bruno-simon.com/>) served as a contrasting form of inspiration. While highly interactive and visually impressive, the portfolio prioritizes creativity and entertainment over clarity and professional presentation. Observing this design helped reinforce my decision to focus on readability, organization, and usability rather than heavy animations or complex interactions, since my goal was to create a portfolio suitable for professional and academic purposes.

While these websites influenced certain design ideas, the final portfolio was adapted to reflect my own technical background and interests in software engineering, AI, and UI/UX design, while maintaining a professional and modern appearance.

5. Portfolio Website Link

The portfolio website was deployed publicly using Netlify to ensure accessibility and ease of review. The website is fully responsive and publicly accessible without authentication.

Portfolio Link:

<https://ajaysuthaharan.netlify.app/>

6. Source Code Repository

The source code for the portfolio website is hosted publicly on GitHub. The repository contains the React-based implementation, reusable components, routing structure, assets, and styling used throughout the project.

GitHub Repository:

<https://github.com/Ajay757/Portfolio.git>

7. Design Choices and Visual Communication

7.1 Colors

A dark color palette was selected to create a professional and modern appearance while reflecting a software and technology-oriented identity. Dark navy and black tones were used for the background, while blue accents were added to emphasize headings, interactive elements, and important information. Muted white and gray tones were used for secondary text to maintain readability and reduce visual clutter.

7.2 Typography

Typography was chosen to support readability and visual hierarchy. Large bold headings were used to emphasize important sections, while smaller muted text was used for descriptions and supporting details. Differences in font size, weight, and spacing helped organize information and improve readability throughout the portfolio.

7.3 Screen Layout and Visual Communication Principles

The portfolio uses a Bento-style card layout to organize content into clear sections while maintaining balance and readability.

Scale: Larger hero text, headings, and featured content were used to emphasize important information.

Visual Hierarchy: Content was organized using spacing, typography, and color emphasis to guide attention toward key sections such as projects, experience, and contact information.

Balance: Different card sizes were used to distribute content evenly across the page and prevent visual overcrowding.

Contrast: Bright text and blue accents were contrasted against dark backgrounds to improve readability and highlight interactive elements.

Overall, the layout was designed to be visually clean, easy to navigate, and professional while reflecting my interests in software engineering, AI, and UI/UX design.

8. Conclusion

This assignment provided an opportunity to design and develop a professional portfolio website using React and Tailwind CSS while applying concepts related to UI design and visual communication. Throughout the project, I focused on creating a clean, modern, and responsive interface that effectively presents technical experience, projects, and personal information.

The assignment also reinforced concepts such as visual hierarchy, contrast, spacing, typography, and layout organization while improving my experience with reusable React components and responsive frontend development. Overall, the final portfolio was designed to function both as a course deliverable and as a professional website that can continue to be expanded in the future.